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Machine Learning-1: Introduction

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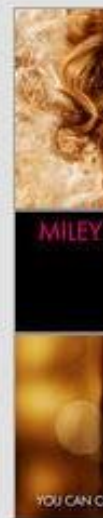
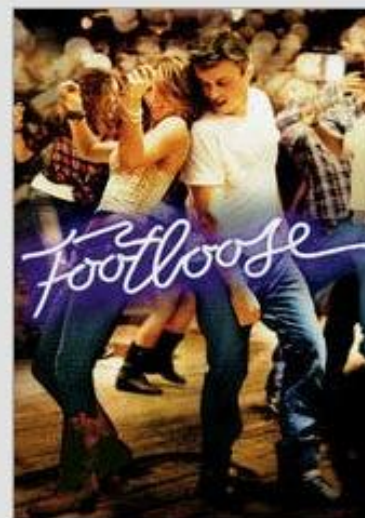




Recently Watched

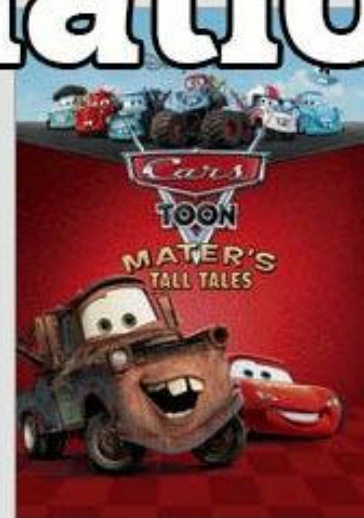
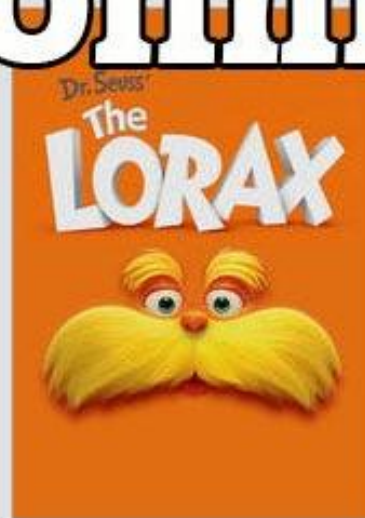


Top 10 for Mark



Netflix Movie

Popular on Netflix



Recommendations





Why do you think?

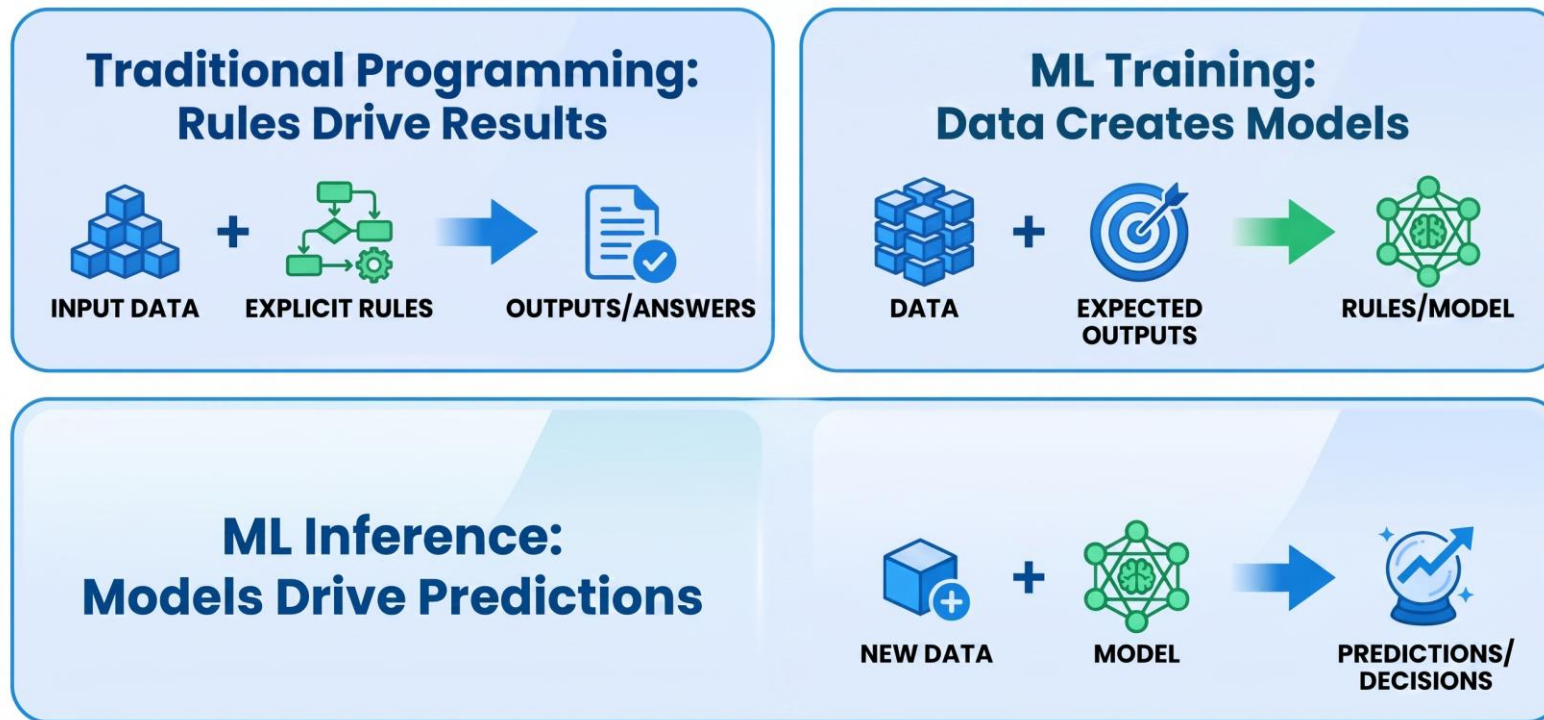
- Some examples are not AI
- Some examples are AI
- ???

When do we lean to AI/ML

- Uncertainty in decision making
- No clearly defined relationship of input and output variables (=> you cannot form rules apriori)
- Too many factors make things confusing
- Scope of utilizing past data to substantiate decision

Difference of ML and Traditional Automation

Programming vs. Machine Learning: The Logic Flip



NotebookLM

What is Artificial Intelligence?

- **Artificial Intelligence** is an exciting scientific discipline that studies how we can make computers exhibit intelligent behavior, e.g. do those things that human beings are good at doing.

Let's now think?

- How can we make computers emulate intelligent human behavior??
- Observe how a human takes a decision
- Encode the decision-making process as rules
- Write a computer program for the rules
- Input the data
- Will it work?

Michael Polanyi (The Tacit Dimension)

- “We know more than what we can tell”
- This means it is extremely difficult to encode the intricacies of human thinking process!
- Hence “Rule-Based” expert systems will not work.
- Small demo video

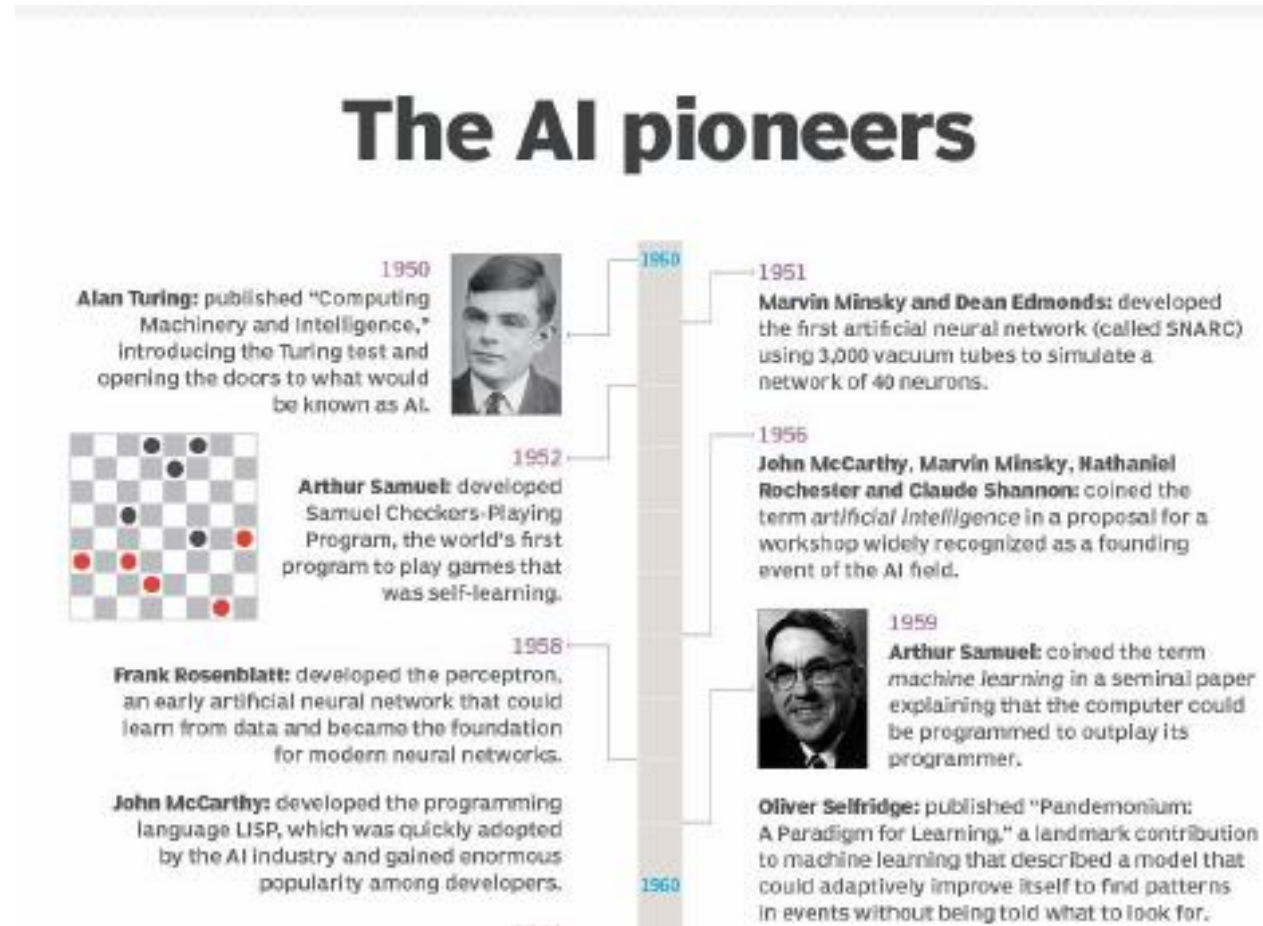
Course Details & Evaluation

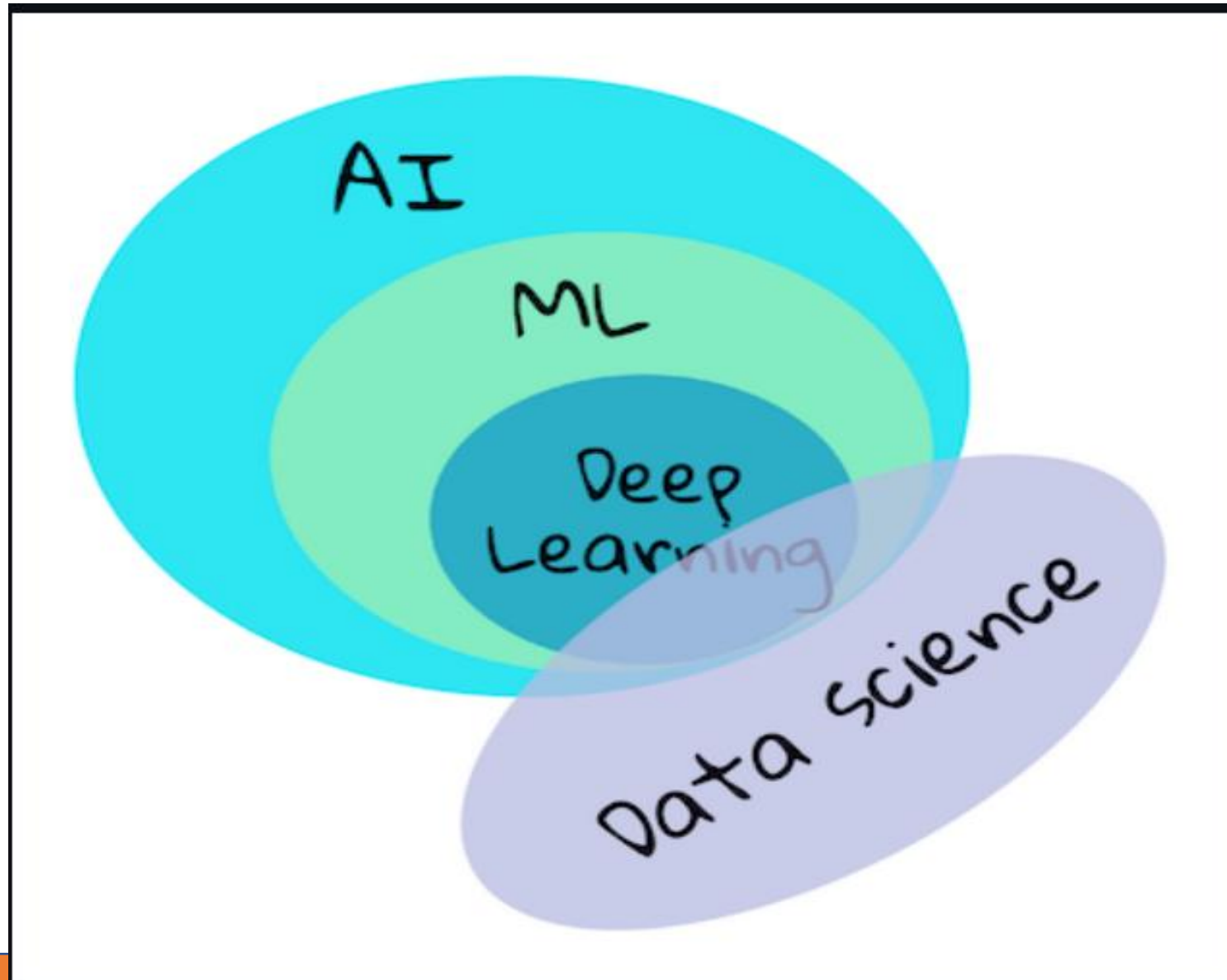
- Course Structure
 - Refer course outline shared in Moodle
- Evaluation
 - Class participation (10%)
 - Mid Term Quiz-1 (20%)
 - Mid Term Quiz-2 (20%)
 - Class Assignments (10%)
 - End Term MCQ Exam (40%)

Important Dates

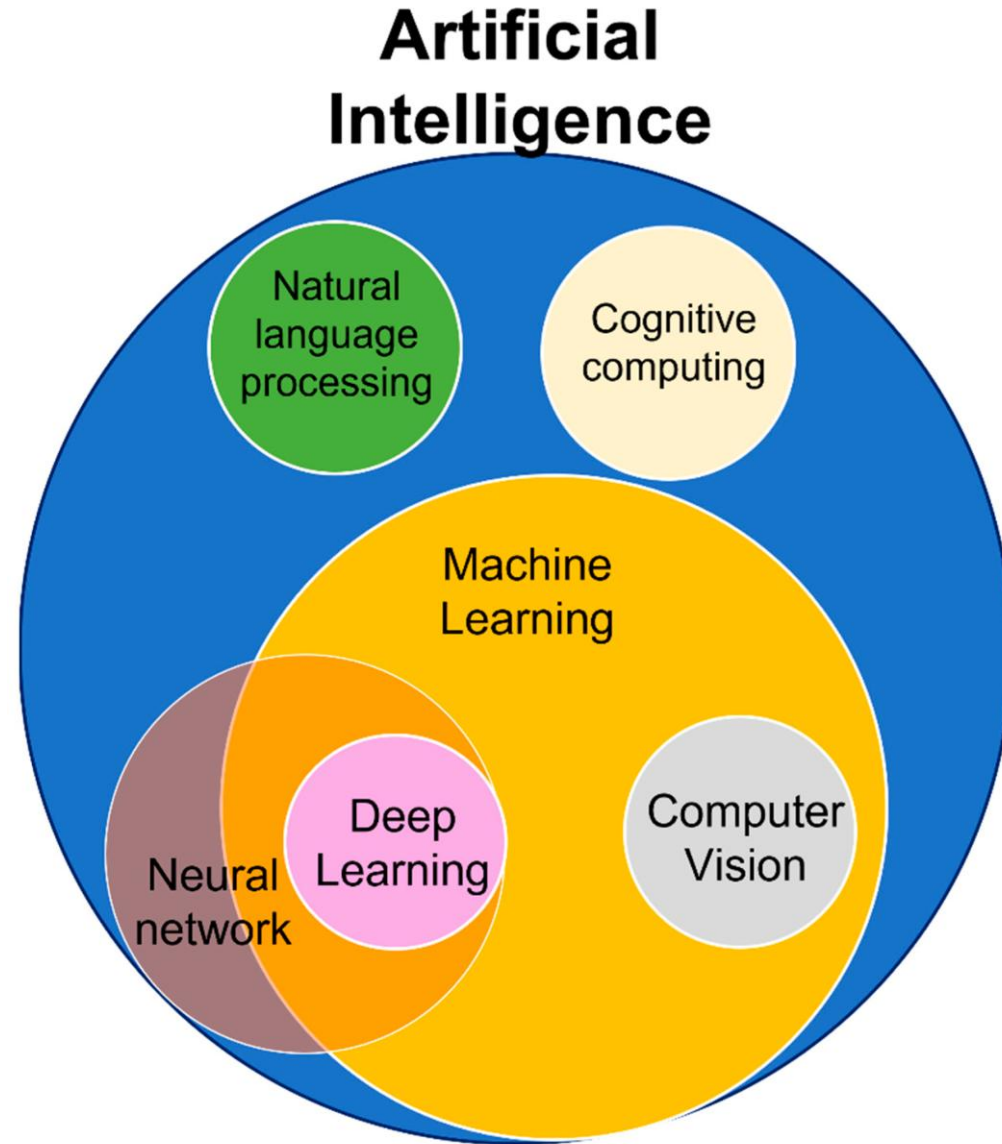
- Mid Term Quiz-1 : **6th Oct**
- Mid Term Quiz-2 : **8th Dec**
- Class Assignment – 1: **29th Sept**
- Class Assignment – 2: **24th Nov**
- End Term Exam : As per EXAM office (MCQ based)
- Class Participation: Based on participation in class discussion and attendance

Evolution of AI/ML





Components of AI



Machine Learning Definition

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E . (Tom Mitchel)
- It's a process of learning

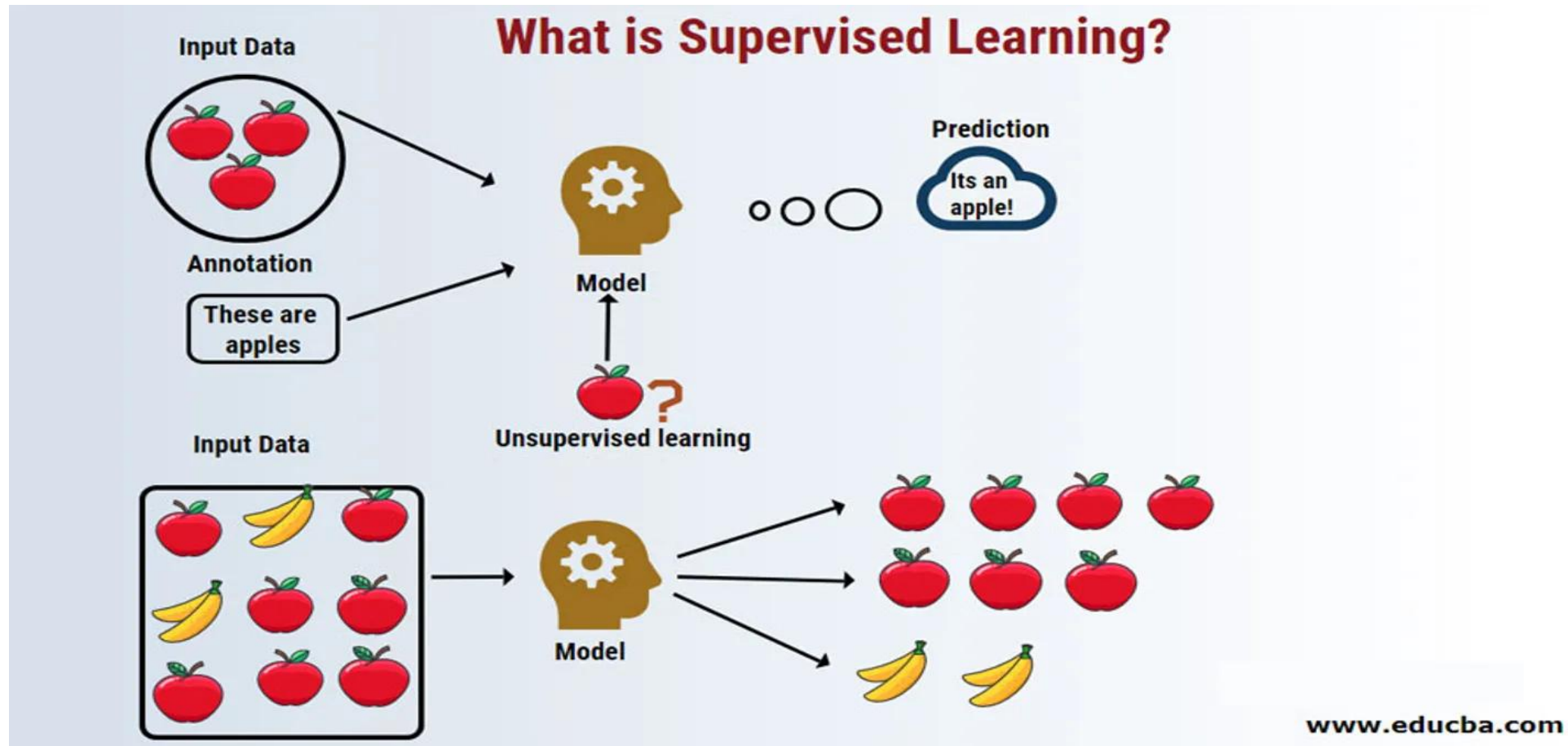
Problem Solving with Machine Learning

- To predict the likelihood of a disease from a patient's medical history
- To leverage weather data to predict weather events
- To understand the sentiment of a text
- To detect fake news and stop the spread of propaganda
- To recommend movies, music, book, dress
- To predict the churn of an employee from a loyalty program
- To predict the price of a house from its attributes
- To predict the price of a share for coming week

Learning Process

- Machine Learning
 - Supervised Learning
 - Unsupervised Learning
 - Semi Supervised Learning
 - Reinforcement Learning

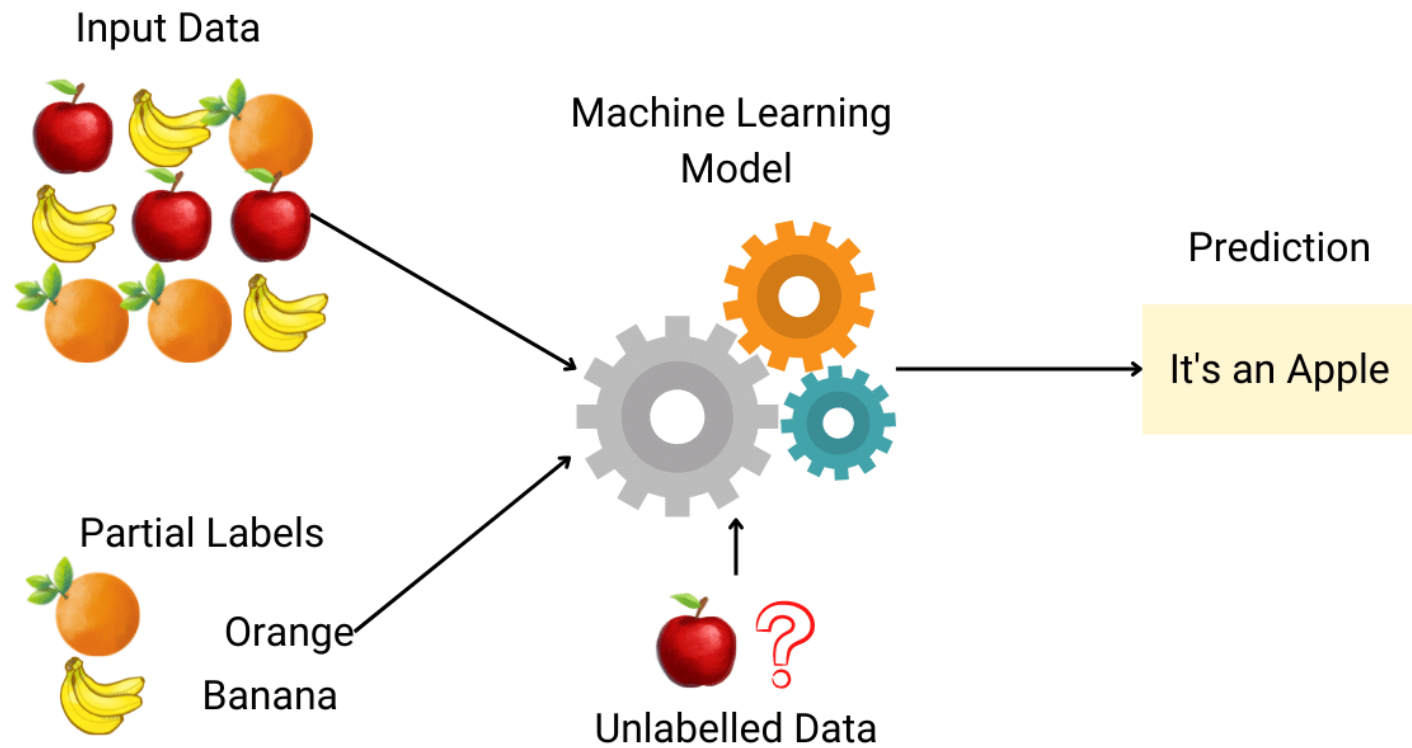
Supervised vs Unsupervised Learning



Difference between Supervised Learning & Unsupervised Learning

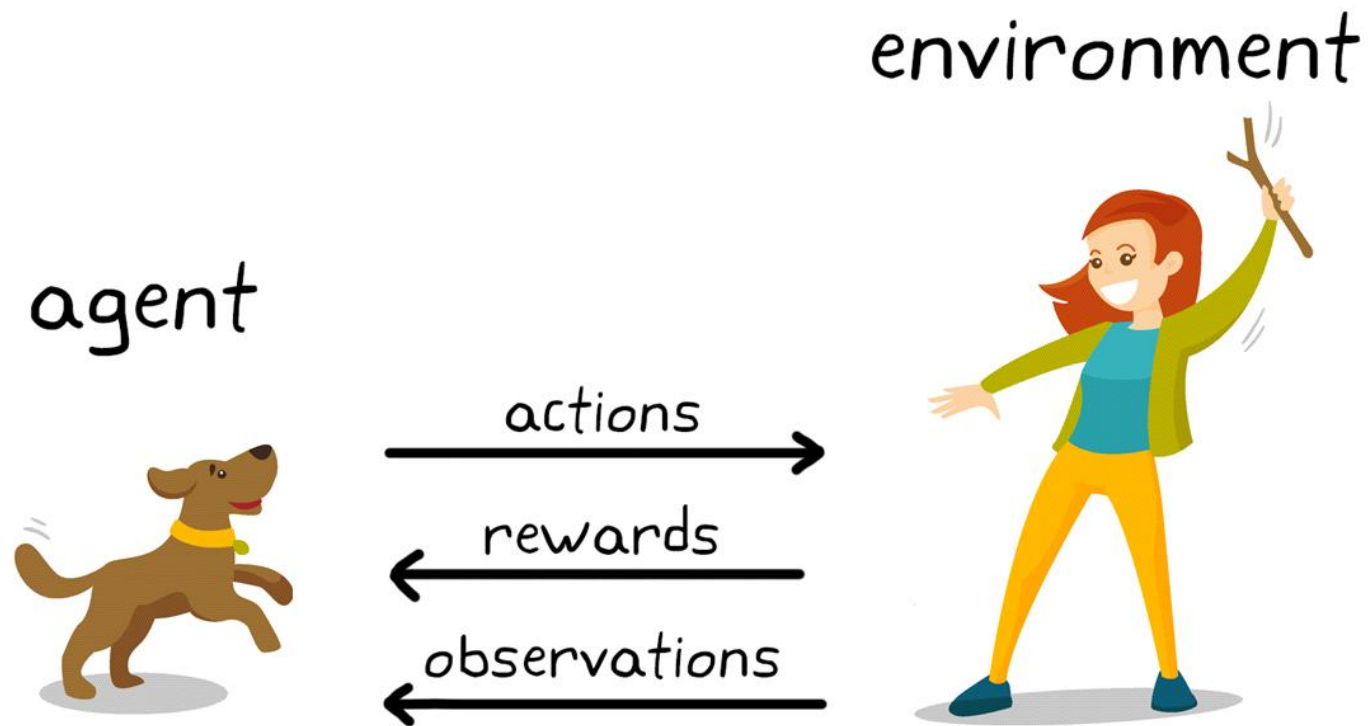
Supervised Learning	Unsupervised Learning
Input data is labelled	Input data is <u>unlabeled</u>
Uses training dataset	Uses just input dataset
Used for prediction	Used for analysis
Classification and regression	Clustering, density estimation and dimensionality reduction

Semi-supervised Learning



enjoyalgorithms.com

Reinforcement Learning



Simulation Demo

- Various types of Machine Learning

Machine Learning (ML) Algorithms

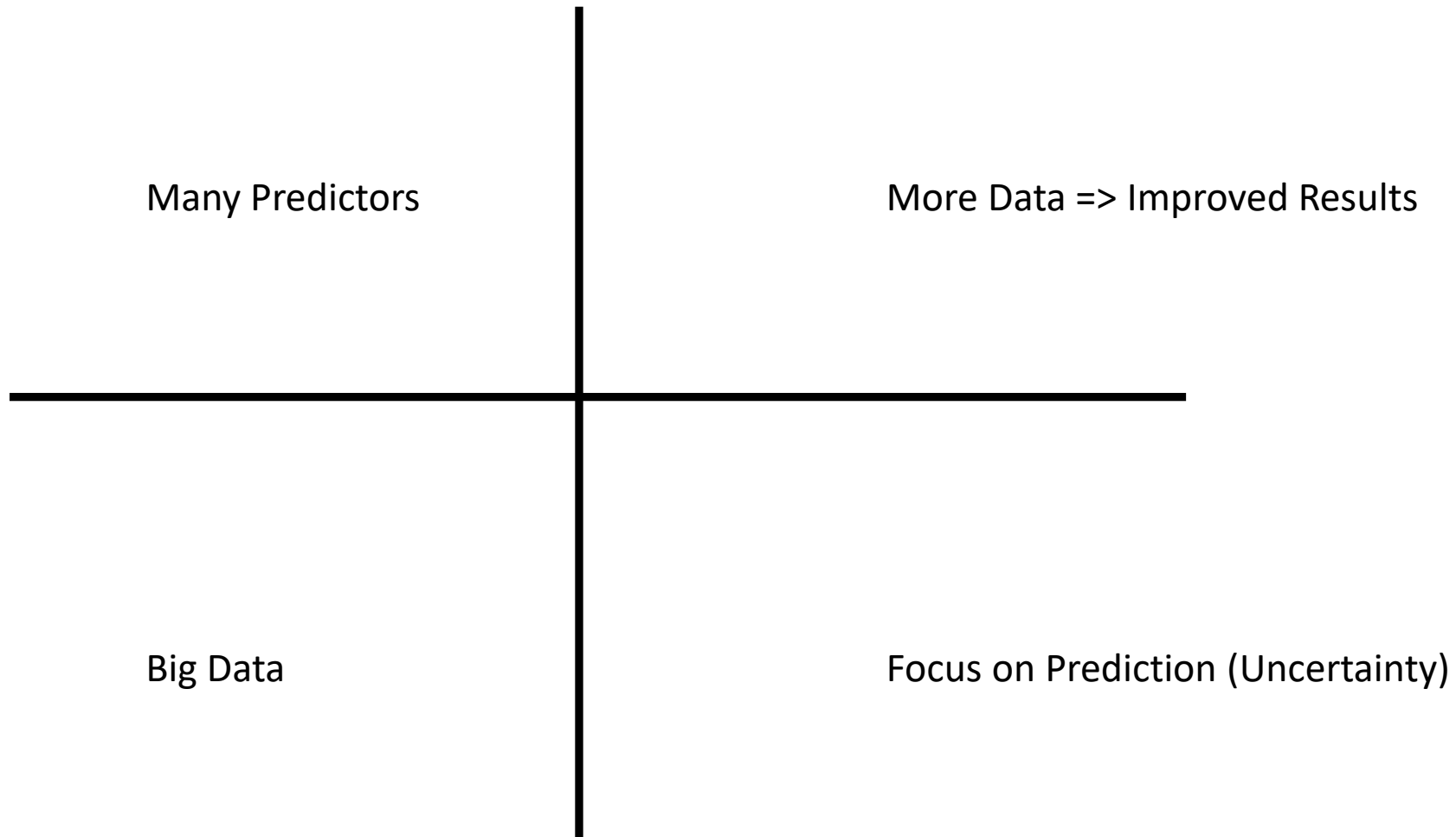
- Multiple Linear Regression(?)
- Regularization with Ridge, Lasso and Elastic Net
- Clustering (K-Means, Spectral, Density-based)
- K Nearest Neighbour (KNN)
- Support Vector Machines
- Market basket analysis, Time series analysis
- Logistic Regression
- Decision Tree (DT), Random Forest (RF), Ensemble models
- Gradient descent, Stochastic gradient descent
- Bayesian Models (Naïve-Bayes, Bayes Net)
- Artificial Neural Network (ANN), CNN, RNN
- Large Language Models (LLM)

What is Machine Learning?

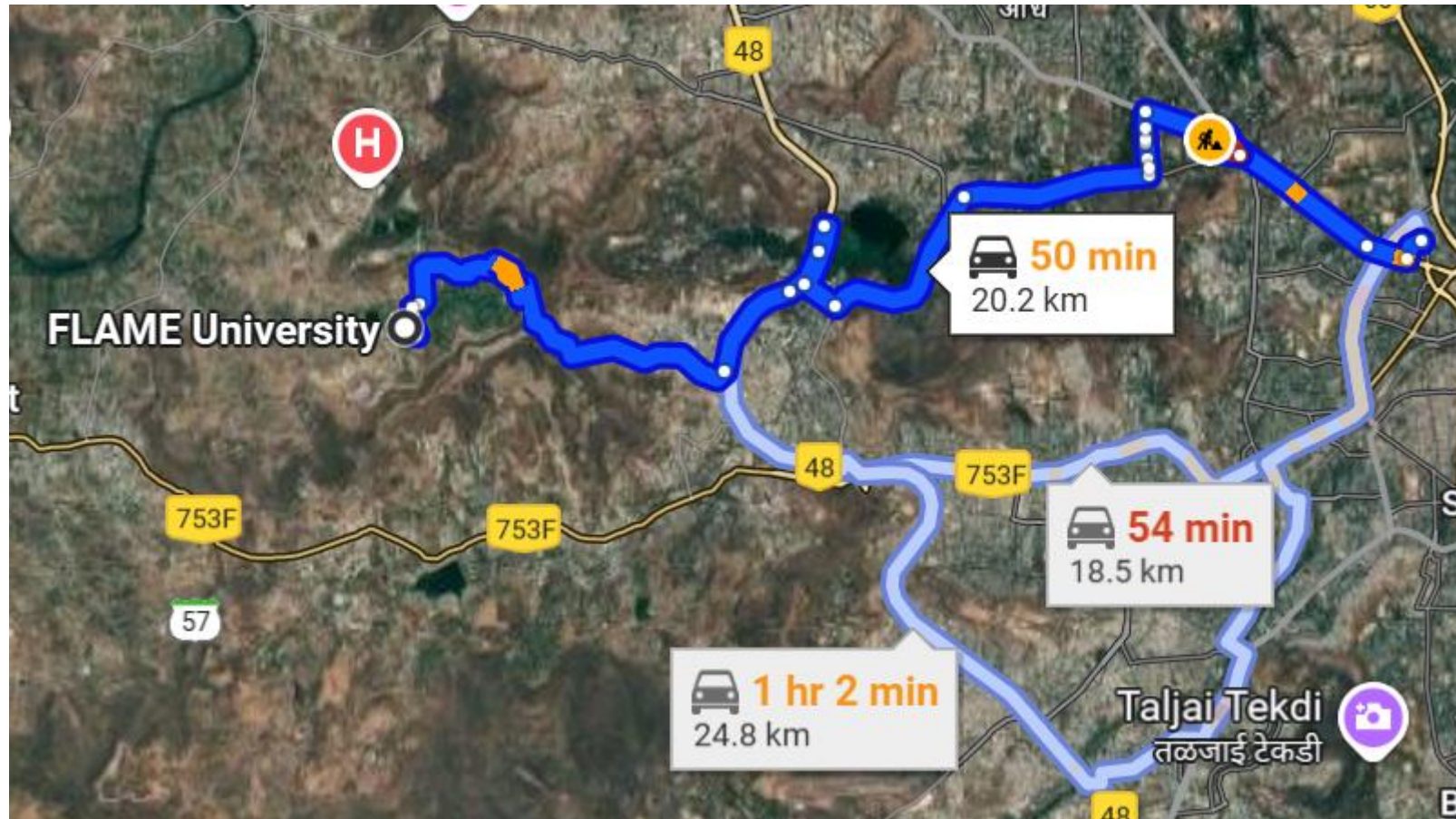
- I create a linear regression model to predict credit score – Is this ML?

When do we say MLR is Machine Learning?

- Big data => Data is large
- Many predictor/explanatory variables, interaction effects
- Focus on Prediction and identifying invisible patterns
- More data improves predictive power with time (like human learning)
- Uncertainty of outcome



Is this ML?



Types of Data

- **Structured Data**
 - Matrix form, rows and columns, Tabular data
 - Rows are “records”, Columns are “attributes”
 - Examples: Database records, Excel datasets
- **Unstructured Data**
 - Not in matrix form
 - Free flowing
 - Examples: Text, Speech, Audio, Video

Challenges of Training Data

- Many sources of truth
- Finding sufficient data points
- Labeling data points
- Selecting important features
- Data cleaning
 - Aggregation
 - Missing values
 - Scaling/Normalization
 - Standardization

Machine Learning Model Creation

- Identify the business problem
- Define input and output of the ML model
- Select appropriate data
- Get to know the data
- Create a model set
- Build models
- Assess models
- Deploy models
- Assess results

ML3: Lesson Goals

- Basic theory of Supervised learning
- Revision of linear algebra and understand its link with ML
- Introduction to Python
- Installing Anaconda/Jupyter
- Basic visualization with Python

Supervised Learning

Supervised Learning (directed data mining, learning with a teacher):

- The variables can be split into two groups:
 - **independent variables** (explanatory variables, inputs, predictors)
 $X = (X_1, \dots, X_p)$;
 - One-dim (or high-dim) **dependent variables** (output, responses) Y
- The objective is to predict Y given values of the input X .

Supervised Learning

- Observed Data (**Training Data set**):
 $(Y_i, X_{i1}, \dots, X_{ip})$ for $i = 1, \dots, n$
- Objective: find a function $h(x_{new}) = h(x_1, \dots, x_p)$ that can predict Y well for any given input $x_{new} = (x_1, \dots, x_p)$ in the **testing data set**.
- Deterministic or randomized relationship?
(many classification tasks in machine learning are deterministic)

Concept of Loss function

Loss function: when $h(X)$ is used to predict Y , the loss is $L(Y, h(X))$.

Two population choices:

- $L(Y, h(X)) = (Y - h(X))^2$ (**squared error loss** for regression)
- $L(Y, h(X)) = I(Y \neq h(X))$ (**0-1 loss** for classification)

Generative vs Predictive ML

- **Generative Models**

- Generative models aim to understand how the data is generated.
- They model the joint probability distribution $P(X,Y)$ over the input data X and output labels Y .
- Generative models can be used to generate new samples from the learned distribution, such as creating new images, text, or even predicting missing parts of data.

- **Examples:**

- Generating realistic images using GANs (Generative Adversarial Networks).
- Producing text using models like GPT (Generative Pre-trained Transformer).
- Modeling the distribution of data in unsupervised learning.

Generative vs Predictive ML

- **Predictive Models (Discriminative Models)**

- Predictive models focus on distinguishing between different classes or predicting outcomes based on input data.
- They model the conditional probability $P(Y|X)$, where Y is the label or outcome you want to predict given the input X .
- The primary goal is to accurately predict the output for new inputs.

- **Examples:**

- Classification models like logistic regression, support vector machines (SVM), or decision trees.
- Regression models that predict a continuous value, like linear regression.
- Neural networks used in supervised learning tasks.

Machine Learning vs Applied Statistics

- ML is a combination of Statistics & Computer Science
 - Tree, Search, Optimization, Heuristics
 - P-value, Hypothesis, Probability, T-test, F-test
- As an ML expert, you need to understand both domains to some extent

Recap: Linear Algebra

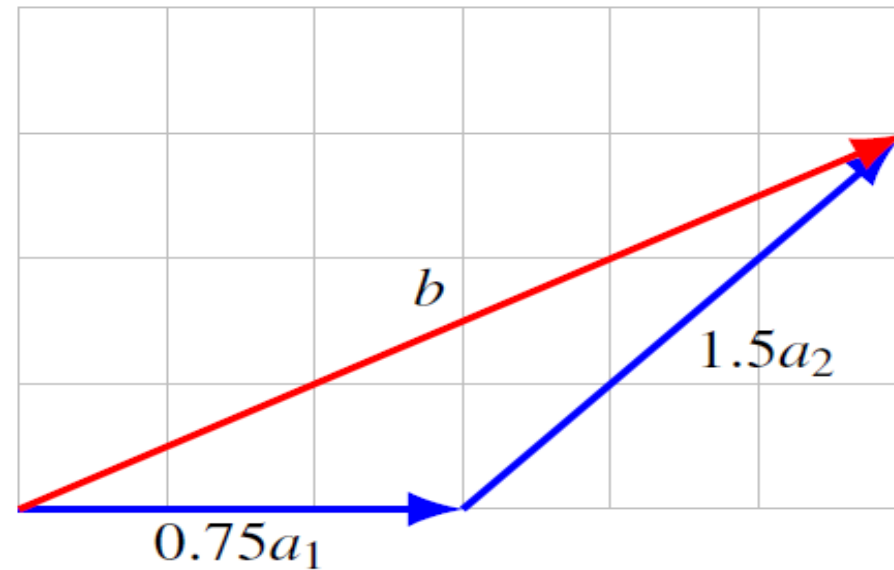
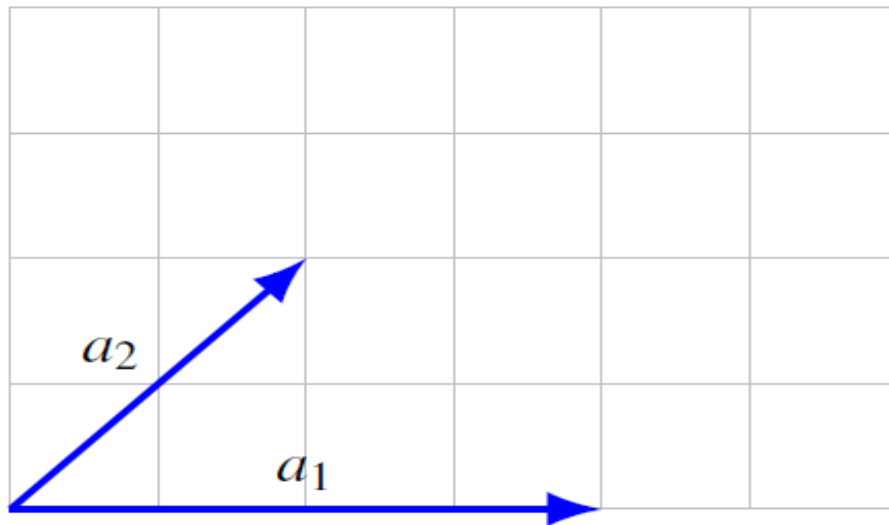
- Vector: row and column vector
- Vector examples
- Vector addition, subtraction, scalar multiplication
- Linear combination of vectors
- Inner product (Dot product)
- Linear transform of a vector
- Python library: numpy

Vector Examples

- ▶ color: (R, G, B)
- ▶ quantities of n different commodities (or resources), *e.g.*, bill of materials
- ▶ portfolio: entries give shares (or \$ value or fraction) held in each of n assets, with negative meaning short positions
- ▶ cash flow: x_i is payment in period i to us
- ▶ audio: x_i is the acoustic pressure at sample time i (sample times are spaced 1/44100 seconds apart)
- ▶ features: x_i is the value of i th *feature* or *attribute* of an entity
- ▶ customer purchase: x_i is the total \$ purchase of product i by a customer over some period
- ▶ word count: x_i is the number of times word i appears in a document

Linear Combination of Vectors

two vectors a_1 and a_2 , and linear combination $b = 0.75a_1 + 1.5a_2$



Inner Product (Dot Product)

- ▶ *inner product (or dot product)* of n -vectors a and b is

$$a^T b = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n$$

- ▶ other notation used: $\langle a, b \rangle$, $\langle a|b \rangle$, (a, b) , $a \cdot b$
- ▶ example:

$$\begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} = (-1)(1) + (2)(0) + (2)(-3) = -7$$

Linear Combination vs Linear Transform

- Let x and y be two vectors in R^n . Then $v = a x + b y$ is a linear combination. Let A be a matrix of dimensions $m \times n$. Then $v = A x$ is a linear transformation from R^n to R^m .
- A linear combination is a sum of (scaled) vectors within a vector space. A linear transformation is a mapping from one vector space to another vector space which is linear.
- In other words, if V and W are two vector spaces, then T from V to W is a linear mapping if it satisfies $T(x + y) = T x + T y$ and $T(a x) = a T(x)$ for all x, y in V and for all scalars a .

Why we need these Linear Algebra techniques in Machine learning?

- **Vectors:** Data representation
- **Dot product:** Used in cosine similarity used in document similarity computation
- **Linear Transformation:** Used in PCA and SVM, for dimension mapping, image mapping, core of ML
- **Linear Combination:** Used in weighted models, most ML models, e.g. LR, Logistic Regression, SVM etc are weighted models

Recap: Matrix

- Matrix
- Matrix transpose
- Matrix inverse
- Matrix multiplication
- Eigenvalue, Eigenvector

Recap: Calculus

- Continuous function
- Derivative of a function
- Find Maxima, Minima of a function
- Random Variables
- Probability Distributions (Uniform, Normal, Poisson, Bernoulli, Binomial)
- Marginal and Conditional Probabilities
- Mean, Variance, Covariance
- Unbiased Estimators
- Maximum

Python notebook

- Linear algebra
 - Matrix basics
 - Data analysis
 - Basic visualization
-
- We will go through the shared notebooks

Who is the creator of Python?



Guido van Rossum - Brief Bio



Here's a brief bio that I send to people who ask me for one. You can also use either of the photos above: the one on the left was taken in 2006, the one on the right in 2019. [Click](#) for larger versions.

Guido van Rossum is the creator of the Python programming language. He grew up in the Netherlands and studied at the University of Amsterdam, where he graduated with a Master's Degree in Mathematics and Computer Science. His first job after college was as a programmer at CWI, where he worked on the ABC language, the Amoeba distributed operating system, and a variety of multimedia projects. During this time he created Python as side project. He then moved to the United States to take a job at a non-profit research lab in Virginia, married a Texan, worked for several other startups, and moved to California. In 2005 he joined Google, where he obtained the rank of Senior Staff Engineer, and in 2013 he started working for Dropbox as a Principal Engineer. In October 2019 he retired. After a short retirement he joined Microsoft as Distinguished Engineer in 2020. Until 2018 he was Python's BDFL (Benevolent Dictator For Life), and he is still deeply involved in the Python community. Guido and his family live in Silicon Valley, where they love hiking, biking and birding.